Lucie Brolon

PhD Student in Applied mathematics

Laboratoire MAP5 (UMR 8145), Université Paris Cité, France

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PROFESSIONAL EXPERIENCE

2023 – Present **PhD in Applied Mathematics, MAP5 Laboratory – Université Paris Cité**

- PhD project on **modeling communication between human T cells and dendritic cells using random forest methods**.
- Supervised by Fabien Crauste and Marie Perrot-Dockès.
- Funded by the French National Research Agency (ANR) through the DCComm Project.

2026 - Present **Temporary Teaching and Research Assistant (ATER) - Université Paris Cité**

- Teaching in the Master's program in Public Health at the Faculty of Pharmacy.
- Practical sessions in **R, Python and SQL**.
- Introduction to supervised and unsupervised learning.

2023 - 2026 **Teaching Assistant (64 hours / year) - IUT de Paris Rives de Seine**

- Statistical programming tutorials in R for second-year undergraduate students.
- Practical sessions on regression using R for first-year undergraduate students.
- Course coordination, including the development of teaching materials and assessments

2021 – 2023 **Research Engineer - INSERM and Institut Pasteur**

- Management and descriptive analysis of data from the French National Health Data System (SNDS).
- Provided **SAS** support to other team members.
- Analyzed healthcare pathways of patients with COVID-19.

2021 **Statistical Biomarker Internship - Sanofi, R&D Department | 6 months**

- Evaluation of different models (ANOVA, ANCOVA, LIMMA) for identifying pharmacodynamic biomarkers in clinical trials.
- Simulation study to assess model performance.

PUBLICATIONS

Preprint:

- ***Linear Modeling, Random Forests and Interpretability: SPLIM, Stable Piecewise Linear Interpretable Model.***
  Lucie Brolon, Fabien Crauste, Marie Perrot-dockès. Preprint HAL, 2026.

Peer-reviewed articles:

- ***Differential impact of the pandemic first wave on chronic respiratory diseases in France***
  Muriel Fartoukh, Lucie Brolon, Marie Al Rahmoun, Alexandre Sabaté-Elabbadi, Didier Guillemot, Christian Brun-Buisson and Laurence Watier. Respiratory Medicine and Research, Volume 90, 2026.
- ***One-year healthcare trajectories after COVID-19 hospitalisation in patients with chronic respiratory diseases: a nationwide study.***

Alexandre Sabaté-Elabbadi, Lucie Brolon, Christian Brun-Buisson, Marie Al Rahmoun, Didier Guillemot, Muriel Fartoukh and Laurence Watier. BMJ Open Respiratory Reserach, Volume 12, Issue 1, 2025.

- ***Trends in hospitalisations for lower respiratory tract infection after the COVID-19 pandemic in adults with chronic respiratory disease.***
  Alexandre Sabaté-Elabbadi, Lucie Brolon, Christian Brun-Buisson, Didier Guillemot, Muriel Fartoukh, Laurence Watier. Journal of Infection, Volume89, Issue5, 2024.

EDUCATION

2020 – 2021	**Research Master's in Public Health** Université de Rennes I, Rennes, France. Specialization in Modeling in Clinical Pharmacology and Epidemiology. Dual degree program between ENSAI and Université de Rennes I.
2018 – 2021	**Engineering Degree, Specialization in Statistics for Life Sciences** ENSAI (National School for Statistics and Information Analysis), Bruz, France.
2016 – 2018	**ENS D2 Preparatory Class in Mathematics and Economics** Lycée Victor et Hélène Basch, Rennes, France. Second-year undergraduate studies in Economics and Management at Université de Rennes I. Preparatory program for the École Normale Supérieure Paris-Saclay (economic track).

CONFERENCE TALKS AND POSTERS

January 2026	MAP5 Modeling, Analysis and Simulation Team Meeting and MAP5 PhD Students' Open Seminar, Paris, France. Oral presentation.
November 2025	SFdS (French Statistical Society) Biostatistics Day, Montpellier, France. Poster presentation.
May 2025	BioInference Conference, Bardonecchia, Italy. Poster presentation.
January 2025	IRMAR PhD Students' Statistics Seminar, Rennes, France. Oral presentation.
November 2024	SFdS Biostatistics Day, Paris, France. Oral presentation.
November 2024	Young Researchers in Applied Mathematics Conference, Lyon, France. Oral presentation.
April 2024	10th SFdS Young Statisticians Meeting, Porquerolles, France. Oral presentation.

SCIENTIFIC OUTREACH AND ENGAGEMENT

2025 – 2026	**Science Outreach.** Introduced classification algorithms to secondary school students at the MAP5 laboratory and EIB Étoile high school.
2024 – 2025	**Math C pour L Workshops.** Participated in workshops co-organized by the French Statistical Society (SFdS), the French Society for Applied and Industrial Mathematics (SMAI), and the French Mathematical Society (SMF), introducing undergraduate women to mathematical research.
2025	**Gender Equality in Science.** Led discussions with two high school classes on gender and scientific careers, around the question "Is becoming a scientist a gendered choice?"
2024 – 2025	**Women in Mathematics Outreach.** Participated in an outreach event at MAP5 as part of the L'Oréal Foundation's *For Girls in Science* program, and regularly took part in online speed-meeting events organized by the *femmes et mathématiques* association for female high school students.